



**K. R. MANGALAM UNIVERSITY**

THE COMPLETE WORLD OF EDUCATION

# Computer Networks (ENCS352)

LAB FILE SUBMITTED TO

**K. R. Mangalam University**

FOR

**Bachelor of Technology**

**in**

**Computer Science and Engineering**

**SUBMITTED BY**

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## Assignment 2: Packet Flow Visualization Using Simulation Mode

### Experiment Objectives:-

Students will use Packet Tracer Simulation Mode to observe how packets move between devices, how switches learn MAC addresses, and how ICMP echo request/reply travels through a small network.

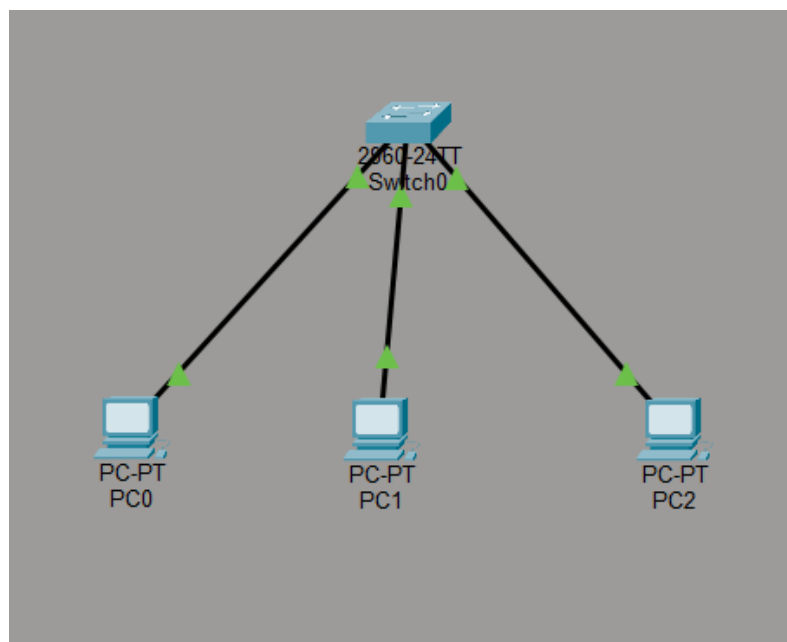
### Learning Outcomes:-

- Use Simulation Mode effectively
- Observe ARP + ICMP behavior
- Understand switch MAC learning behavior
- Interpret packet event list timeline

### Concepts Used:-

- Simulation Mode
- ICMP ping
- ARP resolution
- Switch MAC address table learning

### Task 1: Build a Simple LAN



## Configuration:-

- 1 Switch (2960)
- 3 PCs connected to switch
- IP addresses assigned:

| PC  | IP Address   | Subnet Mask   |
|-----|--------------|---------------|
| PC0 | 192.168.20.1 | 255.255.255.0 |
| PC1 | 192.168.20.2 | 255.255.255.0 |
| PC2 | 192.168.20.3 | 255.255.255.0 |

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.1

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.2

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.20.3

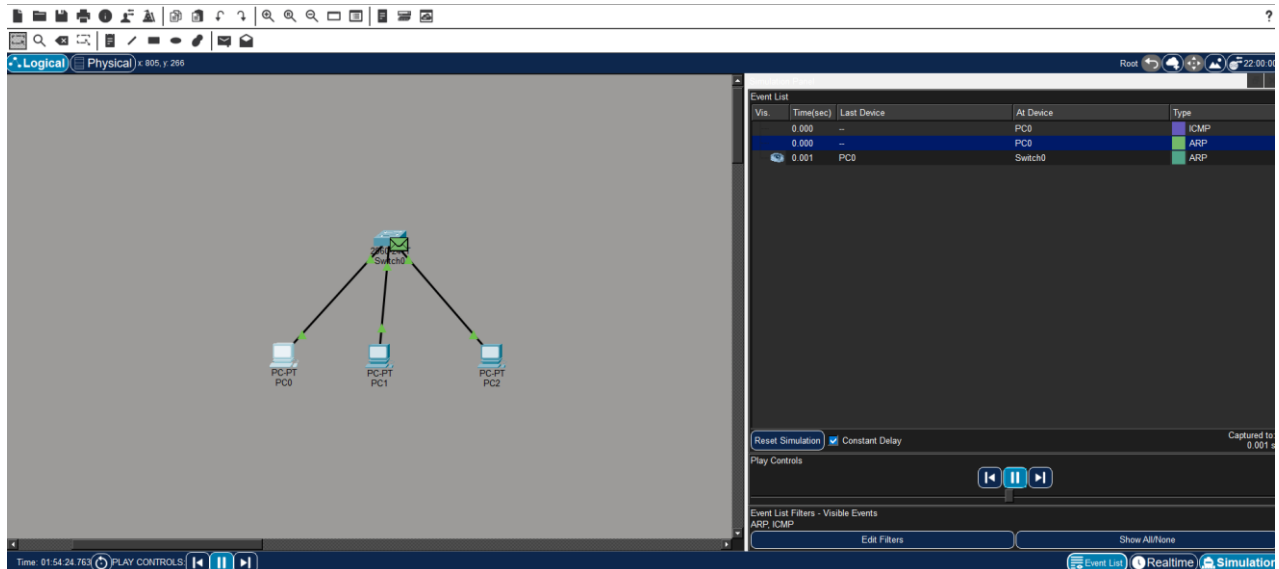
Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

## Task 2: First Ping Observation

- Simulation Mode was enabled (Shift+S or the stopwatch icon in the bottom-right of Packet Tracer).



- The Event List filter was set to show only ARP and ICMP PDUs.



- A ping was issued from PC1 to PC2 via the Command Prompt: ping 192.168.20.2

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.2

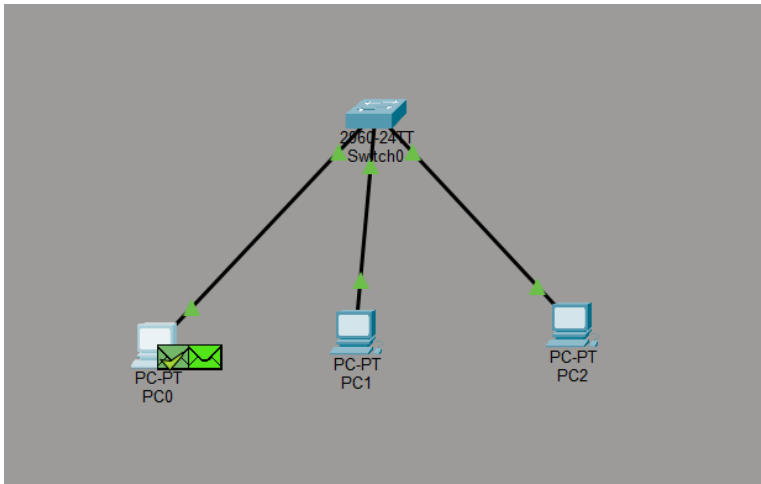
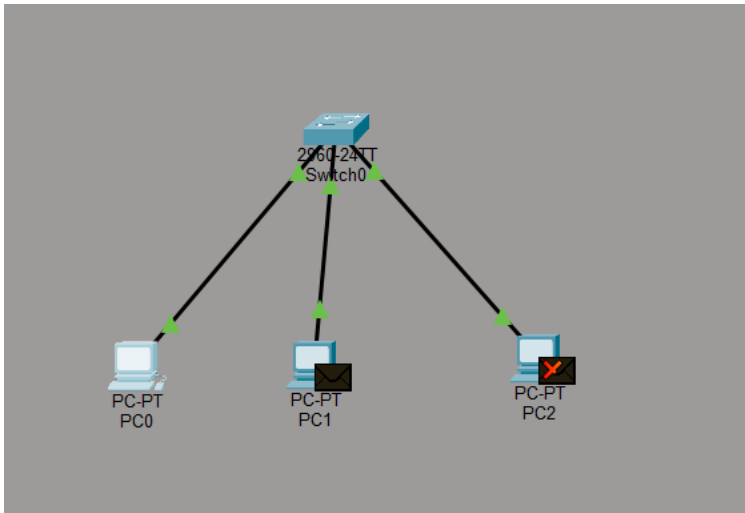
Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=8ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 8ms, Average = 5ms
```

Observed Event Sequence:-

| Simulation Panel |           |             |           |      |
|------------------|-----------|-------------|-----------|------|
| Event List       |           |             |           |      |
| Vis.             | Time(sec) | Last Device | At Device | Type |
|                  | 0.000     | --          | PC0       | ARP  |
|                  | 0.001     | PC0         | Switch0   | ARP  |
|                  | 0.002     | Switch0     | PC1       | ARP  |
|                  | 0.002     | Switch0     | PC2       | ARP  |
|                  | 150.056   | --          | PC0       | ICMP |
|                  | 150.056   | --          | PC0       | ARP  |
|                  | 150.057   | PC0         | Switch0   | ARP  |
|                  | 150.058   | Switch0     | PC1       | ARP  |
|                  | 150.058   | Switch0     | PC2       | ARP  |
|                  | 150.059   | PC1         | Switch0   | ARP  |
|                  | 150.060   | Switch0     | PC0       | ARP  |
|                  | 150.060   | --          | PC0       | ICMP |
|                  | 150.061   | PC0         | Switch0   | ICMP |
|                  | 150.062   | Switch0     | PC1       | ICMP |
|                  | 150.063   | PC1         | Switch0   | ICMP |
|                  | 150.064   | Switch0     | PC0       | ICMP |
|                  | 151.066   | --          | PC0       | ICMP |



### Task 3: Second Ping Observation

Without clearing the ARP cache, a second ping was issued immediately: ping 192.168.20.2. The simulation was re-run from the beginning of the event list for the new set of packets.

```
C:\>ping 192.168.20.2

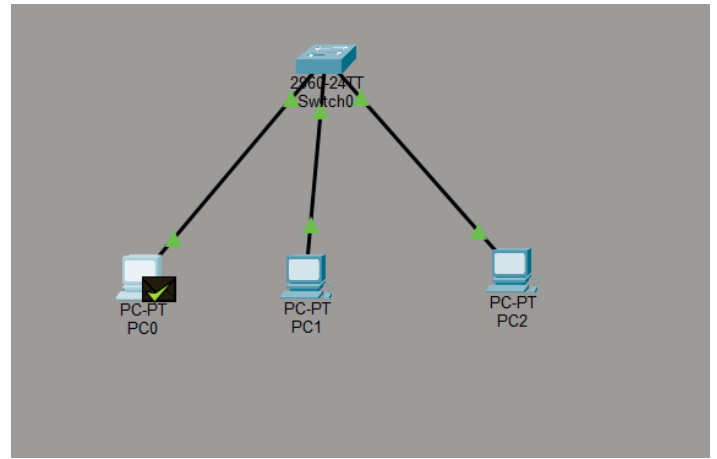
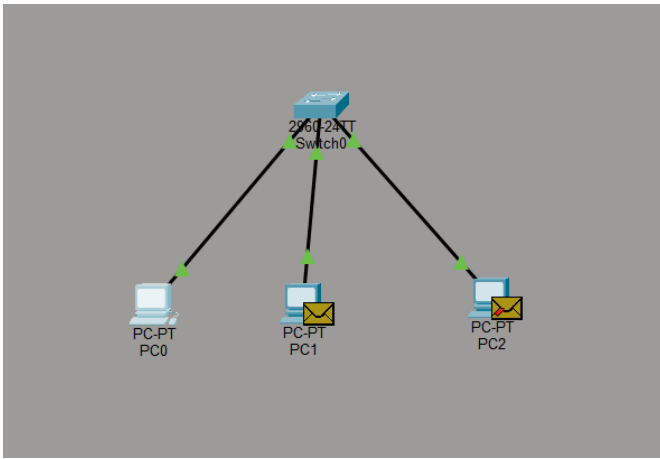
Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128
Reply from 192.168.20.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 4ms, Average = 4ms
```

### Observed Event Sequence:-

| Simulation Panel                                                                    |           |             |           |      |
|-------------------------------------------------------------------------------------|-----------|-------------|-----------|------|
| Event List                                                                          |           |             |           |      |
| Vis.                                                                                | Time(sec) | Last Device | At Device | Type |
|                                                                                     | 0.000     | --          | PC0       | ICMP |
|                                                                                     | 0.001     | PC0         | Switch0   | ICMP |
|                                                                                     | 0.002     | Switch0     | PC1       | ICMP |
|                                                                                     | 0.002     | Switch0     | PC2       | ICMP |
|                                                                                     | 0.003     | PC1         | Switch0   | ICMP |
|                                                                                     | 0.004     | Switch0     | PC0       | ICMP |
|                                                                                     | 1.006     | --          | PC0       | ICMP |
|                                                                                     | 1.007     | PC0         | Switch0   | ICMP |
|                                                                                     | 1.008     | Switch0     | PC1       | ICMP |
|                                                                                     | 1.009     | PC1         | Switch0   | ICMP |
|                                                                                     | 1.010     | Switch0     | PC0       | ICMP |
|                                                                                     | 2.013     | --          | PC0       | ICMP |
|                                                                                     | 2.014     | PC0         | Switch0   | ICMP |
|                                                                                     | 2.015     | Switch0     | PC1       | ICMP |
|                                                                                     | 2.016     | PC1         | Switch0   | ICMP |
|                                                                                     | 2.017     | Switch0     | PC0       | ICMP |
|                                                                                     | 3.021     | --          | PC0       | ICMP |
|  | 3.022     | PC0         | Switch0   | ICMP |



#### Task 4: MAC Table Check

To verify switch learning, the MAC address table was inspected via the switch CLI in Packet Tracer

```
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1    0090.2195.8e29    DYNAMIC Fa0/2
      1    00d0.9749.a3b3    DYNAMIC Fa0/1
Switch#
```

- First Ping vs. Second Ping – Comparison

| Feature       | First Ping  | Second Ping |
|---------------|-------------|-------------|
| ARP Used      | Yes         | No          |
| Speed         | Slow        | Fast        |
| Packets Count | More        | Less        |
| Reason        | MAC unknown | MAC cached  |